

Jared Lemler

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PROFESSIONAL SUMMARY

Backend Engineer with a craftsman's approach, 8 years in electrical systems, now building software systems that are safe, reliable, and maintainable. My electrician background honed my systems thinking. I had to coordinate teams, read complex plans, and build an infrastructure that can't fail. I bring the same patience and troubleshooting discipline from wiring buildings to designing APIs, databases, and containerized services.

Technically grounded in Python/FastAPI and TypeScript/Fastify, with production experience in PostgreSQL, Prisma, Docker. Recently architected VoxPort, a federated podcast platform on the AT Protocol, where I built high throughput data pipelines (1,000 event bounded in-memory queue, 100 event batched persistence, ~95% I/O reduction) and learned to operate under real-world constraints. I work with honest communication, clear ownership, and passion for contributing to teams.

SKILL OVERVIEW

- **Languages:** Python, JavaScript, TypeScript, SQL
- **Databases:** PostgreSQL
- **Data & ETL:** Pandas, Pydantic Validators, Scheduled Jobs (hourly/daily), batch processing, CSV Ingestion, JSON Pipelines
- **DevOps & Tooling:** Docker, Alembic, Git, GitHub, JWT, pytest, Node.js, Sentry observability, OpenTelemetry tracing, structured logging, Axiom
- **Cloud & Infrastructure:** AWS (EC2, Elastic Beanstalk, S3), CI/CD (CodeQL, Gitleaks, Bandit), Cloudflare R2 (content-addressed storage), Hetzner, Neon, Fly.io
- **Core Competencies:** REST API Design, Authentication & Authorization, Data Modeling, Distributed Systems, AT Protocol, WebSocket, Real-time Event Processing
- **AI Tools:** Zed, Devin, Devin Cloud, opencode, Mastra

Frameworks/Libraries: Prisma ORM, FastAPI, Fastify, SQLAlchemy 2.0, Pydantic, Django, React,

PROJECTS

VoxPort (<https://voxport.live>) - Podcast Platform built on the AT Protocol

- **Zero-Downtime AT Protocol Federation:** Built resilient WebSocket consumer surviving 10 reconnection cycles with exponential backoff and cooloff, with bounded memory backpressure (1,000 event cap per worker), processing 6 collections with 100 event batching. Achieved ~95% I/O reduction vs. naive per event persistence.
- **Engagement Analytics Engine:** Built hourly velocity aggregation system tracking engagement across 1 hour to 168 hours (7 day) windows using a weighted scoring algorithm (likes=1x, mentions=5x) for real-time trending detection via hourly scheduled jobs.

- Memory Optimized Cursor Tracking: Eliminated per event DB writes via in-memory cursor with 5-second flush intervals, reducing database round trips by ~95% during high volume ingestion.
- Relational Data Architecture: Designed 26 entity normalized schema with time-series aggregation tables for historical trend analysis.
- Distributed Extraction Worker: Built standalone fly.io worker claiming MediaJob rows via FOR UPDATE SKIP LOCKED with per source host politeness gating (1 concurrent job per host), exponential backoff retry (30s/60s/120s, max 3 attempts), graceful shutdown finishing in-flight jobs, and scale to zero idle behavior. Keeping ffmpeg and outbound fetches out of request serving API.
- Scalable Media Infrastructure: Integrated Cloudflare R2 for content-addressed blob storage, mapping CIDS to AT Protocol records for verifiable media distribution across federated nodes.
- Backfill Migration System: Developed ETL pipeline for migrating legacy content to AT Protocol, supporting bulk series/episode migration with dry run validation and error tracking.
- Zero-Trust Audio Security: Implemented magic-number file validation (not extensions), ffmpeg metadata stripping, and 450MB limits. Blocking spoofed uploads while preserving creator metadata privacy.

doc-lok <https://www.npmjs.com/package/doc-lok> <https://github.com/JL037/doc-lok> (Preprompt context condenser)

- Context Window Optimization: Engineered a TypeScript pre prompt middleware that condenses prompt payloads by resolving remote markdown links into compact cryptographic markers (<lok_id>), drastically cutting context token
- Conditional Caching Pipeline: Built a two phase network verification engine using HTTP HEAD probes and consumption and API latency.If-None-Match ETag validation to bypass full payload downloads for unchanged remote resources
- Zero Dependency Architecture: Authored a lightweight CLI and library relying exclusively on native Node.js APIs, eliminating third-party supply chain vulnerabilities and minimizing cold-start overhead.

Travel Wishlist <https://www.travelwishlist.app> (Location-Based Travel Planner)

- Architecture & Design: Built with FastAPI, PostgreSQL, and Docker. Implemented JWT authentication, Pydantic validation, and a reusable CRUD service layer.
- Weather API Integration: Built city-based weather lookup system integrating external weather APIs with 100ms response times.
- Calendar-Based Itinerary System: Created monthly/agenda calendar views for trip planning with date-based organization of saved cities and travel scheduling.
- Interactive Map Feature: Mapping layer for geographic visualization of saved destinations.

PROFESSIONAL EXPERIENCE

Talk — Backend Developer (Self-Employed)

Pacifica, CA · Jan 2025 – Feb 2026

(Social Connection Platform – Netherlands)

- Developed and deployed a privacy-compliant web app supporting 200+ university students using React, Vite, Tailwind, and FastAPI backend integrations.
- Translated founder vision into technical implementation using Lovable prototypes, delivering full stack solutions.
- Created a QR code to trigger data entries between physical and digital experiences for social experiments, enabling real-time participation and improving data collection efficiency.
- Built automated data pipelines and dashboards converting raw activity into insights, reducing manual reporting effort by 90%.
- Consulted with team and users, which led to decreased onboarding steps and an increase in user retention in live trials.

- Collaborated with founders to align analytics and engagement metrics for research and product direction.
Technologies Used: TypeScript, React, Plausible Analytics, Tailwind, Vite, Cloudflare

Journeyman Electrician 2017-2025